

Foundations of Arithmetic Power Geometry XI: Spectral Gap Stability Under Canonical Bounded Green–Renormalized Metrics

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Abstract

Arithmetic Power Geometry (APG) is a deformation-theoretic framework in which algebraic closure relations are studied under continuous and regularized exponent deformation. Previous volumes established entropy-governed local closure defects, information-geometric stabilization, integrated defect functionals, scale-corrected invariants, discrete prime-chain regularization, conditional APG–Arakelov height coupling, discrepancy–source compatibility, the Spectral–Green Bridge, source concentration, and entropy-controlled spectral energy. The remaining analytic obstruction in the APG spectral-conductor program is the possible collapse of the first positive eigenvalue of the APG Laplacian on compactified modular curves. This paper develops APG XI by introducing a canonical bounded Green–renormalized APG metric and proving polynomial spectral gap stability for that canonical metric. If

$$X = X_0(N)^*(\mathbb{C}), \quad g_{\text{APG}} = e^{2\phi_{\text{APG}}} g_{\text{std}},$$

where g_{std} is a compact Arakelov-compatible reference metric and ϕ_{APG} is the bounded Green–renormalized APG conformal factor, then

$$\text{osc}_X(\phi_{\text{APG}}) \leq C \log \log N.$$

Consequently,

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}$$

whenever the reference spectral gap satisfies a conductor-polynomial lower bound. In the congruence modular-curve setting, this is compatible with the classical Selberg spectral-gap input for the standard hyperbolic model. The paper does not claim a proof of Fermat’s Last Theorem, the abc conjecture, the Szpiro conjecture, the Birch and Swinnerton-Dyer conjecture, Open Problem 15.1, or the full APG–Arakelov Projection Theorem. Its contribution is the spectral-gap stability component of the APG IX–XIII roadmap under a canonical APG metric normalization.

Keywords: Arithmetic Power Geometry; spectral gap; modular curves; Laplacian; Green kernel; entropy; source concentration; Arakelov geometry; Dirichlet energy; spectral stability; conformal metrics.

1 Introduction

Arithmetic Power Geometry (APG) studies algebraic closure under exponent deformation. In the classical arithmetic-geometric setting, one usually fixes polynomial exponents and studies rational or integral points over arithmetic fields. APG reverses part of this viewpoint: it fixes positive baseline coordinates and treats the exponent as a deformation parameter. The first APG volume introduced the local Power-Deformation Space and proved that the leading-order closure defect near the Euclidean target is governed by Shannon entropy (Akhtar, 2026a; Shannon, 1948; Cover and Thomas, 2006). APG II developed stabilized information geometry and coordinate stability (Akhtar, 2026b; Amari, 2016; Nielsen, 2020). APG III introduced the integrated closure-defect functional, APG IV studied scale correction and coordinate-dependent damping, and APG V replaced ill-defined continuous conductor paths by discrete prime-chain regularization (Akhtar, 2026c,d,e).

The later APG program moves toward the APG–Arakelov architecture. APG VI formulated a conditional APG–Arakelov height-coupling program. APG VII identified boundary terms involving cusp widths, vertical components, and Green-energy distortion. APG VIII proved the Spectral–Green Bridge. APG IX proved source concentration and entropy-controlled APG density bounds. APG X combined APG VIII and APG IX into entropy-controlled spectral energy (Akhtar, 2026f,g,h,i,j).

The present paper is the eleventh volume in this sequence. Its purpose is not to solve the entire APG spectral-conductor program. Its purpose is narrower and more precise: to establish a rigorous spectral-gap stability theorem for a canonically normalized APG metric on compactified modular curves. The target inequality in the broader program is

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N. \quad (1)$$

APG IX supplies the source estimate

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 H(W)^2, \quad (2)$$

while APG X gives the entropy–spectral energy bridge

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2. \quad (3)$$

Thus, after APG IX and APG X, the analytic obstacle is the possible collapse of $\lambda_1(\Delta_{\text{APG}})$. APG XI addresses this issue.

The central idea of the present paper is that spectral stability should not be claimed for arbitrary, unregulated APG metrics. A conformal factor produced by solving a Poisson equation can, in principle, develop large oscillation unless a Green-kernel or metric-distortion theorem is available. Rather than hide this difficulty, APG XI introduces a canonical bounded Green–renormalized APG metric. This metric preserves the APG source-potential architecture while enforcing a transparent conductor-scale distortion ceiling. The main theorem proves that this

canonical metric satisfies

$$\mathrm{osc}_X(\phi_{\mathrm{APG}}) \leq C \log \log N, \quad (4)$$

and therefore gives polynomial spectral stability.

The contribution is best understood as a reduction and normalization theorem. APG XI proves that the spectral-gap obstruction is removed within the canonical bounded APG metric model. It does not assert that every possible APG metric has the same property. This distinction is essential for mathematical honesty and for avoiding circular use of Green-kernel estimates.

2 Classical and APG Background

2.1 Modular curves and spectral geometry

Let

$$X = X_0(N)^*(\mathbb{C})$$

be the compactified modular curve. Its analytic geometry is connected to the theory of modular forms, automorphic representations, and Laplace operators on Riemann surfaces (Shimura, 1971; Diamond and Shurman, 2005; Iwaniec, 2002; Sarnak, 1990). The spectral gap of the Laplacian is classically studied through the decomposition of automorphic forms and the theory of congruence subgroups. Selberg’s spectral theory supplies a fundamental lower-bound paradigm for congruence quotients (Selberg, 1965; Buser, 1992; Iwaniec, 2002). Later work refined the spectral-gap technology and connected it to the theory of automorphic representations (Kim and Sarnak, 2003; Sarnak, 2004).

The arithmetic background of modular curves also draws upon deformation theory, modular Galois representations, arithmetic surfaces, and Diophantine geometry. Foundational developments include the study of ℓ -adic representations attached to modular forms, modular deformation theory, Eisenstein ideals, and arithmetic modular schemes (Deligne, 1971; Katz, 1973; Mazur, 1977; Serre, 1987; Diamond, 1996).

Classical arithmetic-geometric references include (Hartshorne, 1977; Borevich and Shafarevich, 1966; Lang, 1983; Vojta, 1987; Hindry and Silverman, 2000; Silverman, 1986, 2007).

These works provide part of the broader arithmetic setting in which the APG VI–XIII program is situated.

Arakelov geometry provides a natural arithmetic-geometric setting for compact Riemann surfaces arising from arithmetic curves (Arakelov, 1974; Faltings, 1984; Soule, 1992). It connects analytic objects such as metrics, Green kernels, and Laplacians with arithmetic intersection theory (Zhang, 1993; Bombieri and Gubler, 2006). These tools are central to APG VI–VII and remain relevant in APG XI because the APG metric is required to be compact and Arakelov-compatible.

2.2 Green kernels and Poisson potentials

For a compact Riemannian surface (X, g) , the positive Laplacian Δ_g has discrete spectrum

$$0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \cdots.$$

For a mean-zero source F , the Poisson problem

$$\Delta_g u = F, \quad \int_X u d\mu_g = 0$$

has a unique smooth solution when F is smooth, and a weak solution under standard Sobolev hypotheses (Aubin, 1982; Gilbarg and Trudinger, 2001; Grigoryan, 2009). The solution may be represented using the mean-zero Green kernel:

$$u(x) = \int_X G_g(x, y) F(y) d\mu_g(y).$$

However, an L^2 -bound on F alone does not automatically give a conductor-scale L^∞ -bound on u . Such a bound requires heat-kernel, Green-kernel, Sobolev, or spectral input. This point motivates the canonical bounded Green-renormalized definition introduced below.

Related spectral and elliptic estimates arise throughout geometric analysis and Riemannian spectral theory (Berger, 2003; Chavel, 1984).

The interaction between elliptic curves, modularity, and Diophantine equations was emphasized in the Frey-curve program and subsequent modularity developments (Frey, 1986; Ribet, 1990; Taylor and Wiles, 1995; Wiles, 1995).

2.3 APG VIII–X as inputs

APG VIII proved that the APG energy satisfies the Spectral–Green Bridge

$$E_{\text{APG}} \leq \lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2. \quad (5)$$

APG IX proved source concentration:

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 H(W)^2, \quad (6)$$

and in particular $\rho_{\text{APG}} \in L^2(X)$, including cusp neighborhoods. APG X combined (5) and (6) to obtain

$$E_{\text{APG}} \leq C_2 \lambda_1(\Delta_{\text{APG}})^{-1} H(W)^2. \quad (7)$$

Therefore APG XI focuses on $\lambda_1(\Delta_{\text{APG}})$.

3 Scope and Non-Claims

The present paper proves spectral-gap stability for the canonical bounded Green–renormalized APG metric. It does not prove spectral stability for arbitrary APG metrics. It does not prove Fermat’s Last Theorem, the abc conjecture, the Szpiro conjecture, the Birch and Swinnerton-Dyer conjecture, Open Problem 15.1, or the full APG–Arakelov Projection Theorem. The unconditional proof of Fermat’s Last Theorem remains the classical modularity and level-lowering argument developed through the Frey-curve framework and completed by (Frey, 1986; Ribet,

1990; Taylor and Wiles, 1995; Wiles, 1995).

The specific APG XI contribution is the following implication:

canonical bounded APG metric $\implies$ conformal oscillation control $\implies$ polynomial spectral stability.

This is the precise spectral input required before APG XII can address conductor and modular-degree dominance.

4 The Canonical Bounded Green–Renormalized APG Metric

4.1 Reference metric

Let g_{std} be a compact Arakelov-compatible reference metric on $X = X_0(N)^*(\mathbb{C})$, with associated area form $d\mu_{\text{std}}$, volume $V_{\text{std}} = \text{Vol}_{\text{std}}(X)$, positive Laplacian Δ_{std} , and first positive eigenvalue $\lambda_1(\Delta_{\text{std}})$. The reference metric is assumed smooth on the compactified surface. In applications one may take a smoothed compact Arakelov metric or a metric uniformly comparable to a compactified hyperbolic model.

4.2 APG source and potential

Let F_{APG} denote the mean-zero APG source fluctuation:

$$F_{\text{APG}} = \rho_{\text{APG}} - \frac{1}{\text{Vol}_{\text{std}}(X)}, \quad \int_X F_{\text{APG}} d\mu_{\text{std}} = 0. \quad (8)$$

When the source is interpreted relative to the APG IX normalization, the same definition may be adjusted by the equivalent volume normalization; this does not affect the present spectral comparison argument.

Let u_{APG} be the mean-zero Poisson potential

$$\Delta_{\text{std}} u_{\text{APG}} = F_{\text{APG}}, \quad \int_X u_{\text{APG}} d\mu_{\text{std}} = 0. \quad (9)$$

APG IX provides the source-side bound

$$\|F_{\text{APG}}\|_{L^2(\text{std})} \leq C_s H(W), \quad (10)$$

for a structural constant $C_s > 0$. The entropy of the two-point APG weight vector $W = (w_a, w_b)$ satisfies

$$0 \leq H(W) \leq \log 2. \quad (11)$$

Remark 4.1. *The potential u_{APG} may not have a conductor-uniform L^∞ bound from (10) alone. Therefore APG XI does not assume such a bound. Instead, the canonical conformal factor is constructed through a bounded renormalization of u_{APG} .*

Canonical Spectral Metric Principle. The objective of APG XI is not to establish spectral stability for every admissible APG metric. Instead, APG XI identifies a canonical spectral representative within the APG conformal class.

The bounded Green–renormalized metric is therefore regarded as the canonical APG spectral metric. It preserves the APG source-potential architecture while eliminating uncontrolled conformal distortion. All spectral quantities in APG XI are computed relative to this canonical metric.

Consequently, the spectral stability theorem established below should be interpreted as a canonical-metric theorem rather than a statement about arbitrary APG conformal deformations.

4.3 Definition of the canonical metric

Definition 4.2 (Canonical bounded Green–renormalized conformal factor). *For $N \geq 3$, define*

$$\phi_{\text{APG}}(x) = \frac{\log \log N}{1 + \log \log N} \cdot \frac{u_{\text{APG}}(x)}{1 + \|u_{\text{APG}}\|_{L^\infty(X)}}. \quad (12)$$

The canonical bounded Green–renormalized APG metric is

$$g_{\text{APG}} = e^{2\phi_{\text{APG}}} g_{\text{std}}. \quad (13)$$

This definition uses the APG Poisson potential, but it prevents uncontrolled metric distortion.

The factor $(1 + \|u_{\text{APG}}\|_\infty)^{-1}$ is a bounded renormalization; the factor $\log \log N / (1 + \log \log N)$ sets the conductor-scale distortion ceiling.

5 Conformal Oscillation Control

Theorem 5.1 (APG conformal oscillation bound). *For the canonical bounded Green–renormalized conformal factor ϕ_{APG} defined in (12),*

$$\text{osc}_X(\phi_{\text{APG}}) := \sup_X \phi_{\text{APG}} - \inf_X \phi_{\text{APG}} \leq 2 \log \log N. \quad (14)$$

Proof. By definition,

$$\left| \frac{u_{\text{APG}}(x)}{1 + \|u_{\text{APG}}\|_{L^\infty(X)}} \right| \leq 1$$

for every $x \in X$. Hence

$$|\phi_{\text{APG}}(x)| \leq \frac{\log \log N}{1 + \log \log N} < \log \log N.$$

Therefore,

$$\sup_X \phi_{\text{APG}} \leq \log \log N, \quad \inf_X \phi_{\text{APG}} \geq -\log \log N.$$

Subtracting gives

$$\text{osc}_X(\phi_{\text{APG}}) \leq 2 \log \log N.$$

□

Corollary 5.2 (Metric distortion bound). *The canonical APG metric satisfies*

$$(\log N)^{-2} g_{\text{std}} \leq g_{\text{APG}} \leq (\log N)^2 g_{\text{std}}.$$

Proof. By Theorem 5.1,

$$|\phi_{\text{APG}}(x)| < \log \log N.$$

Therefore

$$e^{-2 \log \log N} \leq e^{2\phi_{\text{APG}}} \leq e^{2 \log \log N}.$$

Since

$$e^{2 \log \log N} = (\log N)^2,$$

we obtain

$$(\log N)^{-2} \leq e^{2\phi_{\text{APG}}} \leq (\log N)^2.$$

Multiplying by g_{std} yields the result. □

6 Dirichlet Energy and Measure Comparison

The next step is to translate metric distortion into spectral distortion. Because X is a Riemann surface and $g_{\text{APG}} = e^{2\phi_{\text{APG}}} g_{\text{std}}$, the Dirichlet energy is conformally invariant.

Lemma 6.1 (Dirichlet energy invariance). *For every smooth function $f : X \rightarrow \mathbb{R}$,*

$$\int_X |\nabla f|_{\text{APG}}^2 d\mu_{\text{APG}} = \int_X |\nabla f|_{\text{std}}^2 d\mu_{\text{std}}. \quad (15)$$

Proof. Locally write $g_{\text{APG}} = e^{2\phi_{\text{APG}}} g_{\text{std}}$. Then

$$|\nabla f|_{\text{APG}}^2 = e^{-2\phi_{\text{APG}}} |\nabla f|_{\text{std}}^2, \quad d\mu_{\text{APG}} = e^{2\phi_{\text{APG}}} d\mu_{\text{std}}.$$

Multiplying gives

$$|\nabla f|_{\text{APG}}^2 d\mu_{\text{APG}} = |\nabla f|_{\text{std}}^2 d\mu_{\text{std}}.$$

Integration over X proves the claim. □

Lemma 6.2 (Variance comparison). *Let*

$$K_N = (\log N)^2.$$

Then, for every smooth function $f : X \rightarrow \mathbb{R}$,

$$\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{APG}} \leq K_N \inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{std}}. \quad (16)$$

Proof. By Corollary 5.2, the canonical APG metric satisfies

$$(\log N)^{-2} g_{\text{std}} \leq g_{\text{APG}} \leq (\log N)^2 g_{\text{std}}.$$

Since X has real dimension two, the corresponding area forms satisfy

$$d\mu_{\text{APG}} \leq K_N d\mu_{\text{std}}.$$

Therefore, for every $a \in \mathbb{R}$,

$$\int_X |f - a|^2 d\mu_{\text{APG}} \leq K_N \int_X |f - a|^2 d\mu_{\text{std}}.$$

Taking the infimum over all $a \in \mathbb{R}$ yields

$$\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{APG}} \leq K_N \inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{std}}.$$

This is exactly (16). □

7 Spectral Gap Stability

The first positive eigenvalue of the positive Laplacian can be characterized by the Rayleigh quotient

$$\lambda_1(\Delta_g) = \inf_{f \neq \text{const}} \frac{\int_X |\nabla f|_g^2 d\mu_g}{\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_g}. \quad (17)$$

This formulation avoids technical complications caused by different mean-zero conditions for different measures.

Theorem 7.1 (Spectral comparison for the canonical APG metric). *For the canonical bounded Green-renormalized APG metric,*

$$\lambda_1(\Delta_{\text{APG}}) \geq (\log N)^{-2} \lambda_1(\Delta_{\text{std}}). \quad (18)$$

Proof. For any nonconstant smooth function f , Lemma 6.1 gives

$$\int_X |\nabla f|_{\text{APG}}^2 d\mu_{\text{APG}} = \int_X |\nabla f|_{\text{std}}^2 d\mu_{\text{std}}.$$

Lemma 6.2 gives

$$\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{APG}} \leq (\log N)^2 \inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{std}}.$$

Therefore

$$\frac{\int_X |\nabla f|_{\text{APG}}^2 d\mu_{\text{APG}}}{\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{APG}}} \geq (\log N)^{-2} \frac{\int_X |\nabla f|_{\text{std}}^2 d\mu_{\text{std}}}{\inf_{a \in \mathbb{R}} \int_X |f - a|^2 d\mu_{\text{std}}}.$$

Taking the infimum over all nonconstant smooth f gives

$$\lambda_1(\Delta_{\text{APG}}) \geq (\log N)^{-2} \lambda_1(\Delta_{\text{std}}).$$

□

Assumption 7.2 (Reference spectral lower bound). *The reference metric satisfies a conductor-polynomial spectral lower bound: there exist constants $c_0 > 0$ and $B \geq 0$ such that*

$$\lambda_1(\Delta_{\text{std}}) \geq \frac{c_0}{(\log N)^B} \quad (19)$$

for all sufficiently large N .

Theorem 7.3 (APG XI polynomial spectral stability). *Under Assumption 7.2, the canonical bounded Green-renormalized APG metric satisfies*

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A} \quad (20)$$

for constants $c > 0$ and $A > 0$. More precisely, one may take

$$A = B + 2, \quad c = c_0.$$

Proof. Combining Theorem 7.1 with (19) yields

$$\lambda_1(\Delta_{\text{APG}}) \geq (\log N)^{-2} \lambda_1(\Delta_{\text{std}}) \geq (\log N)^{-2} \frac{c_0}{(\log N)^B}.$$

Thus

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c_0}{(\log N)^{B+2}}.$$

Set $A = B + 2$ and $c = c_0$. □

Corollary 7.4 (Uniform reference gap case). *If $\lambda_1(\Delta_{\text{std}}) \geq c_0 > 0$, then*

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c_0}{(\log N)^2}. \quad (21)$$

Proof. This is Theorem 7.3 with $B = 0$. □

8 Spectral Stability Invariant

Definition 8.1 (APG XI spectral stability invariant). *Let A be an exponent for which Theorem 7.3 holds. Define*

$$\Sigma_{\text{stab}}(N) = \lambda_1(\Delta_{\text{APG}})(\log N)^A. \quad (22)$$

Corollary 8.2. *For the canonical bounded Green-renormalized APG metric,*

$$\Sigma_{\text{stab}}(N) \geq c > 0. \quad (23)$$

Proof. This follows immediately by multiplying (20) by $(\log N)^A$. $\square$

The invariant Σ_{stab} records whether APG spectral geometry remains stable at conductor scale. It is the APG XI analogue of the APG X spectral entropy ratio.

9 Connection with APG XII and APG XIII

APG IX gives the entropy-controlled source estimate

$$\left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_1 H(W)^2.$$

Theorem 7.3 gives

$$\lambda_1(\Delta_{\text{APG}})^{-1} \leq C_2 (\log N)^A.$$

Multiplying yields

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C_3 H(W)^2 (\log N)^A. \quad (24)$$

Therefore APG XII is reduced to proving the conductor/modular-degree dominance estimate

$$H(W)^2 (\log N)^A \leq C_4 \deg(\pi) \log N. \quad (25)$$

If APG XII proves (25), then (24) implies

$$\lambda_1(\Delta_{\text{APG}})^{-1} \left\| \rho_{\text{APG}} - \frac{1}{\text{Vol}(X)} \right\|_{L^2(X)}^2 \leq C \deg(\pi) \log N, \quad (26)$$

which is the spectral-conductor inequality targeted by Open Problem 15.1.

Thus APG XI does not complete the full program. It supplies the spectral-gap component required for APG XII and APG XIII.

10 Numerical Illustration

The main theorem has the form

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}.$$

If the reference spectral gap is uniform and the canonical metric exponent is $A = 2$, then for illustration one obtains

$$\lambda_1(\Delta_{\text{APG}}) \gtrsim \frac{c}{(\log N)^2}.$$

For example, if

$$N = 10^6,$$

then

$$\log N = \log(10^6) \approx 13.8155,$$

and therefore

$$(\log N)^2 \approx 190.87.$$

Hence

$$\lambda_1(\Delta_{\text{APG}}) \gtrsim \frac{c}{190.87} \approx 5.24 \times 10^{-3}c.$$

This numerical example is illustrative only. It is not a proof of any arithmetic inequality. Its purpose is to demonstrate the conductor-polynomial decay scale predicted by the APG XI spectral stability theorem.

11 Discussion

The central danger in APG XI is circularity. One might try to prove conformal oscillation by asserting a Green-kernel estimate of the form

$$\|\Delta_{\text{std}}^{-1}F\|_{L^\infty} \leq C \log \log N \|F\|_{L^2}.$$

But such an estimate is not a consequence of APG IX alone. It requires independent spectral or heat-kernel input. If used without proof, it merely assumes a strong form of the desired spectral stability. The canonical bounded Green–renormalized metric avoids this circularity. It uses the APG Poisson potential, but it normalizes the conformal factor by construction so that metric distortion is bounded at conductor scale.

This method has both strength and limitation. The strength is that the resulting spectral theorem is fully rigorous for the canonical metric. The limitation is that it does not establish spectral stability for every conceivable APG metric. The appropriate conclusion is therefore not “all APG metrics are spectrally stable,” but rather:

the canonical bounded APG metric is spectrally stable at polynomial conductor scale.

This is sufficient as a milestone if subsequent APG volumes explicitly adopt this canonical metric as the APG spectral metric.

12 Conclusion

APG XI proves polynomial spectral gap stability for the canonical bounded Green–renormalized APG metric on compactified modular curves. The proof proceeds through four steps:

- (i) define the canonical conformal factor from the APG Poisson potential using bounded Green renormalization;

- (ii) prove the conformal oscillation estimate $\text{osc}_X(\phi_{\text{APG}}) \leq C \log \log N$;
- (iii) derive logarithmic metric comparison between g_{APG} and g_{std} ;
- (iv) transfer the reference spectral lower bound to the APG Laplacian by Rayleigh quotient comparison.

The resulting theorem is

$$\lambda_1(\Delta_{\text{APG}}) \geq \frac{c}{(\log N)^A}.$$

This supplies the spectral-gap input needed by the APG IX–XIII roadmap. The next step, APG XII, must prove conductor and modular-degree dominance:

$$H(W)^2(\log N)^A \leq C \deg(\pi) \log N.$$

Acknowledgment of Scope

This manuscript is a framework and reduction paper within Arithmetic Power Geometry. It does not claim to replace the classical modular proof of Fermat’s Last Theorem, nor does it claim proofs of the abc conjecture, Szpiro conjecture, BSD conjecture, Open Problem 15.1, or the full APG–Arakelov Projection Theorem. Its contribution is the canonical-metric spectral stability theorem required for the next APG milestone.

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